Turnkey 16S rRNA Metagenomics Analysis Pipeline

End-to-End Workflow: Quality Control -> Taxonomic Profiling -> Diversity Statistics -> Functional Annotation

1. Raw Sequencing Inputs

Cohort: BioProject PRJNA1088906

Paired-End FASTQ Reads

Illumina NovaSeq / MiSeq

* File: data/fastq/*_1.fastq.gz, *_2.fastq.gz
* 17 samples (72k-184k reads/sample)
* 16S rRNA V3-V4 hypervariable region

Clinical Metadata

Patient Cohort Stratification

* File: metadata.tsv
* Cohorts: Healthy (n=4) vs. UC (n=13)
* Variables: SampleID, Group, SRA Accession

Reference DB & Indexing

01_setup.sh Automated Build

Greengenes 16S Database

kraken2-build --special greengenes

* Reference: GG 16S rRNA sequences
* Lineage: NCBI + Greengenes taxonomy
* Files: hash.k2d, opts.k2d, taxo.k2d

Bracken Read-Length Index

bracken-build -l 150 -t 8

* Parameter: Read length = 150 bp
* Precomputed k-mer distribution
* File: database150mers.kmer_distrib

FASTQ

K2 DB

150mer Dist

2. Taxonomic Profiling & Matrix

02_classify_pipeline.sh & 03_downstream_analysis.sh

Kraken2 Exact k-mer Matching

kraken2 --confidence 0.0 --paired

* Algorithm: Lowest Common Ancestor (LCA)
* Multi-threaded execution (GNU Parallel)
* Output: *.kraken & *.report (98.4-99.9% classified)

*.report

Bracken Bayesian Redistribution

bracken -r 150 -l S -t 4

* Algorithm: Bayesian probability re-estimation
* Resolves ambiguous genus/family reads to species
* Output: *.bracken abundance files

*.bracken

Merged Species Abundance Matrix

03_downstream_analysis.sh (pandas outer join)

* File: results/taxonomy/species_abundance_matrix.txt
* Dimensions: 290 species x 17 clinical samples
* Relative abundance fractions (sum = 1.0)

Matrix QC

Sequencing QC & Rarefaction

scripts/qc_depth_analysis.py

* Reads/sample depth boxplot (median: 117k reads)
* Classification efficiency distribution (>98.4%)
* In-silico rarefaction curves (Shannon & Richness)
* Confirms sampling saturation plateau

Raw FastQ Amplicon Sequences (VSEARCH OTU Clustering)

3. Interactive Exploration

03_krona_visualization.sh

Radial Multi-Rank Krona Charts

ktImportText (Hierarchical Pie)

* Interactive HTML: results/krona/all_samples_krona.html
* Full multi-level hierarchy: Domain -> Phylum -> Species

Report Tree

4. Diversity & Statistical Suite

04_diversity_analysis.sh & statistical scripts

Multi-Level Taxonomic Profiles

scripts/plot_multilevel_taxa.py

* Stacked barplots: Phylum & Genus compositions
* Group-mean comparative composition (Healthy vs UC)
* Top-25 dominant species hierarchical clustered heatmap

Taxa Matrix

Alpha & Beta Diversity Suite

scripts/diversity_analysis.py

* Alpha diversity: Shannon, Simpson, Observed, Evenness
* Beta diversity: Bray-Curtis dissimilarity distance matrix
* PCoA 2D Ordination (Axis 1: 28.4%, Axis 2: 19.8%)

Abundance TSV

Species Counts

Species Abundance Matrix

Hypothesis Testing & Networks

differential_abundance.py & network_analysis.py

* PERMANOVA: R2 = 0.156, F = 2.772, p = 0.016*
* Biomarker Volcano & Forest plots (BH-FDR < 0.1)
* Microbial co-occurrence network (Spearman |r| >= 0.6)

Clinical Metadata (Healthy vs. UC Cohorts)

5. Functional Annotation

05_functional_picrust2.sh & profiling.py

OTU Clustering & Centroids

prepare_picrust2.py (VSEARCH)

* Dereplication & sequence clustering at 97% identity
* Centroid representative fasta: rep_seqs.fasta
* Sample OTU abundance table: otu_table.tsv
* Formats amplicons for phylogenetic placement

rep_seqs & otu_table

PICRUSt2 Metabolic Inference

picrust2_pipeline.py (MetaCyc/KEGG)

* Phylogenetic hidden state prediction (hsp.py)
* Enzyme Commission (EC) numbers predicted
* KEGG Orthology (KO) functional gene families
* MetaCyc pathway abundance: path_abun_unstrat.tsv
* Deliverables: Top-20 pathway barplots & heatmaps

MetaCyc & Pathways

Curated Microbial Pathways

scripts/functional_profiling.py

* Short-Chain Fatty Acid (SCFA) metabolism:
  - Acetate synthesis (EC 2.3.1.8 / ackA)
  - Propionate production (acrylate pathway)
  - Butyrate synthesis (buk / but pathway)
* Mucus degradation (Akkermansia muciniphila signature)
* Lactate fermentation & primary carbon cycling
* Deliverable: Curated functional pathway comparative heatmap